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Remagnetization of Cretaceous forearc strata on Santa Margarita and Magdalena Islands, Baja California Sur: Implications for northward transport along the California margin

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Abstract. Paleomagnetic data for two sections of Cretaceous forearc strata with different structural attitudes on Santa Margarita and Magdalena Islands in Baja California Sur, Mexico, indicate that these rocks have been remagnetized, probably during the late Cenozoic. The in situ paleomagnetic directions, however, are similar to data from other Cretaceous rocks on peninsular California with unexpectedly shallow inclinations and easterly declinations. These data have been interpreted as indicating either northward tectonic transport (10°-15° of latitude) and clockwise rotation (>20°) or compaction shallowing of magnetic inclinations in sedimentary rocks combined with southwestward tilting of plutonic rocks. The available paleomagnetic data for Cretaceous forearc strata in southern and Baja California can be divided into three groups: (1) sections with normal-polarity magnetizations that fail fold tests and are remagnetized, (2) sections with normal-polarity magnetizations with no or inconclusive fold tests that may or may not be remagnetized, and (3) sections with both normal- and reversed-polarity intervals where pervasive remagnetization has not occurred. Other rocks of the Mesozoic Great Valley Group, Coast Range ophiolite, and Franciscan Complex in California also have secondary magnetizations with directions similar to younger geomagnetic field directions. Although these widespread remagnetizations could have variable local causes, we propose regional burial and uplift, related to changes in subduction parameters, as a possible explanation. Two episodes of remagnetization are apparent: one in the Late Cretaceous and a second in the late Cenozoic. On the other hand, the unremagnetized and apparently reliable data from sedimentary and plutonic rocks on the Baja Peninsula consistently indicate northward translation (14°± 3°) and clockwise rotation (29°± 8°) with respect to North America since the Late Cretaceous.

1. Introduction

Paleomagnetic data for Cretaceous forearc strata and coeval plutonic rocks in southern and Baja California have been interpreted as indicating northward displacement of this terrane 10°-15° in latitude, relative to stable North America, during Late Cretaceous and early Tertiary time [*Hagstrum and Filmer*, 1990; *Lund and Bottjer*, 1991]. In addition to laboratory examinations of paleomagnetic stability, this conclusion is based on successful fold and reversal tests by data for sedimentary rocks [*Fry et al.*, 1985; *Morris et al.*, 1986], the apparent lack of alteration in both rock types [*Teissere and Beck*, 1973; *Filmer and Kirschvink*, 1989], and similar results from sedimentary and plutonic rocks [*Hagstrum et al.*, 1985]. Moreover, plate motion models indicate that oblique convergence between the North American plate and oceanic plates to the west could have facilitated northward displacement of terranes along the continental margin during Late Cretaceous and early Tertiary time [*Engebretson et al.*, 1985; *Stock and Molnar*, 1988]. Modern analogs of such transport include the southern edge of Sumatra along the Semangko fault system [*Jarrard*, 1986] and the Baja Peninsula itself along the San Andreas fault system [*Beck*, 1986].

A tectonic interpretation, implying significant northward transport of the Baja Peninsula prior to opening of the Gulf of California, has been challenged by workers who infer geologic ties between Paleozoic rocks of peninsular California and mainland Mexico [*Gastil et al.*, 1991] and note the difficulty in establishing such ties with the truncated margin of southern Mexico [*Böhnel et al.*, 1992; *Schaaf et al.*, 1995]. *Butler et al.* [1991] suggested an alternate explanation of the paleomagnetic data in which the shallow paleomagnetic inclinations were caused by burial compaction within the sedimentary rocks, and by regional tilting of the plutonic rocks. The "puzzling coincidence" referred to by *Butler et al.* [1991] is that the compaction effect has equaled the tilting effect. Another coincidence, however, is the similarity between the paleomagnetic poles for Cretaceous rocks of Baja California and the late Cenozoic (Miocene and younger) geomagnetic pole, which may indicate later remagnetization. As previously mentioned, field stability tests and other evidence have indicated that many of these rocks are not remagnetized, but suitable tests are often lacking. Thus, presently there are three possible interpretations, or combinations thereof, of the paleomagnetic data from southern and Baja California: (1) tectonic transport, (2) regional tilting and compaction shallowing, and (3) remagnetization.

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Cases for the first two interpretations have been previously made [*Beck*, 1991; *Butler et al.*, 1991]; here we examine evidence for remagnetization in these rocks, particularly, sedimentary rocks in peninsular California. First, we present new data from Cretaceous forearc strata on Magdalena and Santa Margarita Islands in Baja California Sur (Figure 1) indicating that these rocks have been remagnetized, probably since Miocene time. Next, we discuss paleomagnetic data from sedimentary and other rocks in Alta California (Coast Range ophiolite, Franciscan Complex, Great Valley Sequence) that also have secondary magnetizations, and suggest that remagnetization along the California margin during the latest Mesozoic and Cenozoic was possibly caused by episodes of regional burial and subsequent uplift. Finally, we propose that the simplest interpretation of the most reliable data from the Baja Peninsula is northward transport and clockwise rotation of these rocks since the Late Cretaceous.

2. Geologic Setting in Mexico

Mesozoic oceanic rocks, including Cretaceous strata that are the subject of this study, crop out extensively on Cedros Island and the Vizcaino Peninsula in Baja California [*Patterson*, 1984; *Smith et al.*, 1993], and on Magdalena and Santa Margarita Islands in Baja California Sur [*Sedlock*, 1993, 1996]. These oceanic rocks are separated from Paleozoic and Mesozoic continental rocks to the east by an inferred major fault boundary beneath Cenozoic sedimentary rocks and the Pacific Ocean [*Sedlock et al.*, 1993]. Unmetamorphosed Cretaceous strata unconformably overlie Late Triassic to Late Jurassic island arc and ophiolitic rocks that, in turn, structurally overlie coherent blueschists along shallowly dipping normal fault systems [*Sedlock*, 1988, 1993, 1996]. The Cretaceous strata include terrigenous sandstone, shale, and conglomerate that probably accumulated as turbidites in a forearc basin or basins. In the Cedros-Vizcaino area they are currently assigned to the Albian-Eocene Valle Group [*Smith et al.*, 1993; D. Smith, unpublished data, 1993].

Turbidites in the Magdalena-Santa Margarita area (Figure 1), here informally referred to as the Magdalena Group, strongly resemble the Valle Group, but crop out less extensively and have not been studied in detail. Like the Valle Group, the Magdalena Group shows evidence of terrigenous provenance and deposition in deep-sea fan systems. Magdalena Group strata crop out on southern Magdalena Island and in several small, isolated areas on Santa Margarita Island. For this study, paleomagnetic samples were collected from the two thickest exposures at the eastern end of the major east-west arroyo on southern Magdalena Island (MG1) and in the sea cliff of Cerro Colorado on northern Santa Margarita Island (MG2).

At site MG1, Magdalena Group strata within a fault-bounded wedge form a roughly 400-m-thick section of interbedded sandstone, shale, and laterally discontinuous conglomerate. Shale beds about 90 m above the base of the section contain early Aptian foraminifera (R. Sedlock, unpublished data, 1993), indicating that at least part of the Magdalena Group is older than the Valle Group. Most contacts with adjacent Mesozoic units are faults, but shale locally conformably overlies and grades downward into tuffaceous siltstone of a magmatic arc complex of probable Late Jurassic to Early Cretaceous age. At site MG2, Magdalena Group strata underlying Cerro Colorado form a fault-bounded <100-m-thick section of interbedded sandstone and shale. No age-distinctive microfossils were collected from this area, so biostratigraphic correlation with the section on Magdalena Island is not possible. However, the base of the section lies concordantly on volcaniclastic and pebbly sandstones of an undated magmatic arc complex. This lithologic similarity to the section 10 km to the west on Magdalena Island may indicate that the Cerro Colorado strata are also, at least in part, Aptian in age.

Cretaceous sedimentary rocks in western Baja California have a simple deformational history, inferred from structural, stratigraphic, and geochronologic evidence [*Sedlock*, 1996; 1998]. With other Mesozoic units in western Baja they underwent major brittle extension in an active forearc setting beginning at about 100 Ma and continuing into the Paleogene. Synsubduction extension probably ceased by 40-30 Ma, but comparatively minor faulting and tilting associated with transform motion between the Pacific and North American plates may have occurred since the middle Miocene.

3. Methods and Results

Oriented samples for paleomagnetic analysis were collected using a portable core drill at the two sites on Magdalena and Santa Margarita Islands (Figure 1). Forty samples were collected at site MG1, and forty-eight samples were collected at site MG2, with each sampled zone spanning approximately 200 m of section in the least weathered exposures available. Numerous attitude measurements (strike/dip) were made within the sections and, because of their similarity, were averaged to make overall corrections for each of the two sites (Table 1). The remanent magnetization of individual specimens, cut from the core samples, was measured with a cryogenic magnetometer. Specimens were progressively demagnetized with standard alternating field (af) and thermal procedures to 100 mT and 550°C, respectively. Lines representing characteristic magnetization directions were fitted to the demagnetization data using a least squares method based on principal component analysis [*Kirschvink*, 1980], and the statistics of *Fisher* [1953] were used in analyzing the site mean directions (Table 1).

In general, a single component of magnetization was found in these rocks (Figure 2); the natural remanent magnetization (NRM) directions as well as the characteristic line directions are well grouped. In a few of the samples, however, the vector endpoints on the demagnetization plots move toward a point away from the origin (Figure 2b) and show more complex behavior at high demagnetization steps, implying the presence of a second high-coercivity component of magnetization. No consistent direction for this component could be determined. These results are similar to those for the Albian to Santonian Valle Group of central Baja [*Hagstrum et al.*, 1985] and for Campanian turbidites (Cachuma Formation) sampled at Agua Caliente Canyon in southern California [*Hagstrum et al.*, 1996]. Thermal demagnetizations for the Valle and Magdalena Groups and southern California turbidites also show blocking temperatures between 100° and 550°C, although often with unstable results above 400°-550°C. Isothermal remanent magnetization experiments for the Cachuma Formation indicate

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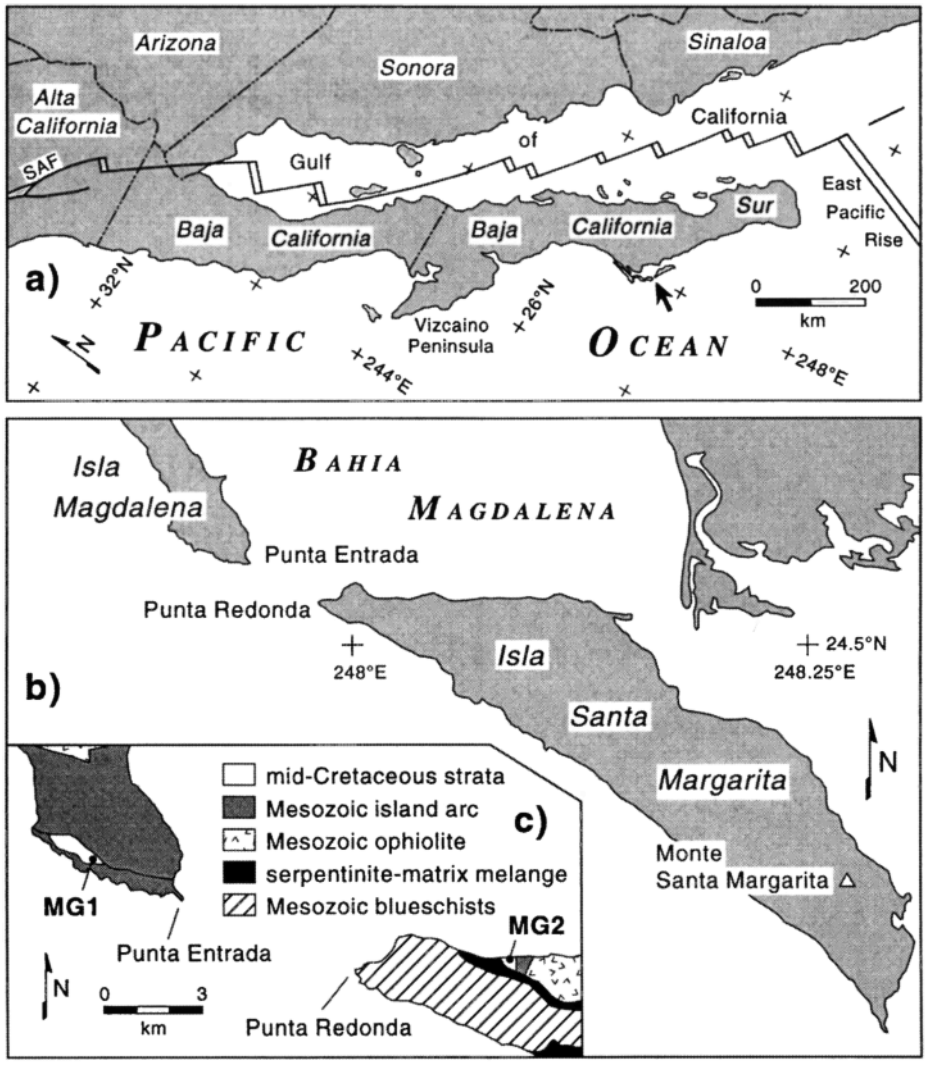


Figure 1. Map of (a) the Baja California Peninsula in southwestern North America and (b) the Bahia Magdalena area on the western coast of Baja California Sur, Mexico. (c) Sampling sites (MG1 and MG2) are located near the southern and western ends of Magdalena and Santa Margarita Islands, near Puntas Entrada and Redonda, respectively.

Table 1. Paleomagnetic Data for Mid-Cretaceous Strata on Santa Margarita and Magdalena Islands, Baja California Sur, Mexico

Site	I	D	N	R	k	α_{95}	λ	ϕ
MG1	46.7	1.5	39	38.6130	98	2.3	86.3	268.8
MG1c	54.1	311.6					47.1	183.3
MG2	67.3	0.0	48	47.0749	51	2.9	64.4	248.1
MG2c	31.3	40.1					51.9	341.6

Here, I and D are inclination and declination of paleomagnetic directions in degrees, c means corrected for structural tilt (MG1: strike = 147°, dip = 39°W; MG2: strike = 334°, dip = 46°E); N is number of samples analyzed; R is vector sum of N unit vectors; k is the concentration parameter [*Fisher*, 1953]; α_{95} is the radius of 95% confidence in degrees. λ and ϕ are latitude and east longitude of virtual geomagnetic pole in degrees. Sampling localities: MG1, 24.6°N, 247.9°E; MG2, 24.5°N, 248.1°E.

that the magnetic mineral in these rocks is predominantly Ti-poor magnetite [*Hagstrum et al.*, 1996].

The sample- and site-mean directions for sites MG1 and MG2 are to the north with moderate to steep inclinations (Figure 3); normal polarity results would be expected in rocks deposited during the Cretaceous Normal Polarity Superchron (Aptian through Santonian; *Harland et al.* [1982]). Upon correction for structural tilt, however, MG1 moves to the west with steeper inclination, and MG2 moves to the east with shallower inclination. This divergence after correction (fold test failure) implies that the magnetization of the Magdalena Group was acquired after major tilting of Late Cretaceous to Paleogene age. The statistically significant (95% confidence level; *McFadden and Lowes* [1981]) in situ difference between the MG1 and MG2 site mean directions, however, may indicate significant postremagnetization deformation of these rocks in the late Cenozoic.

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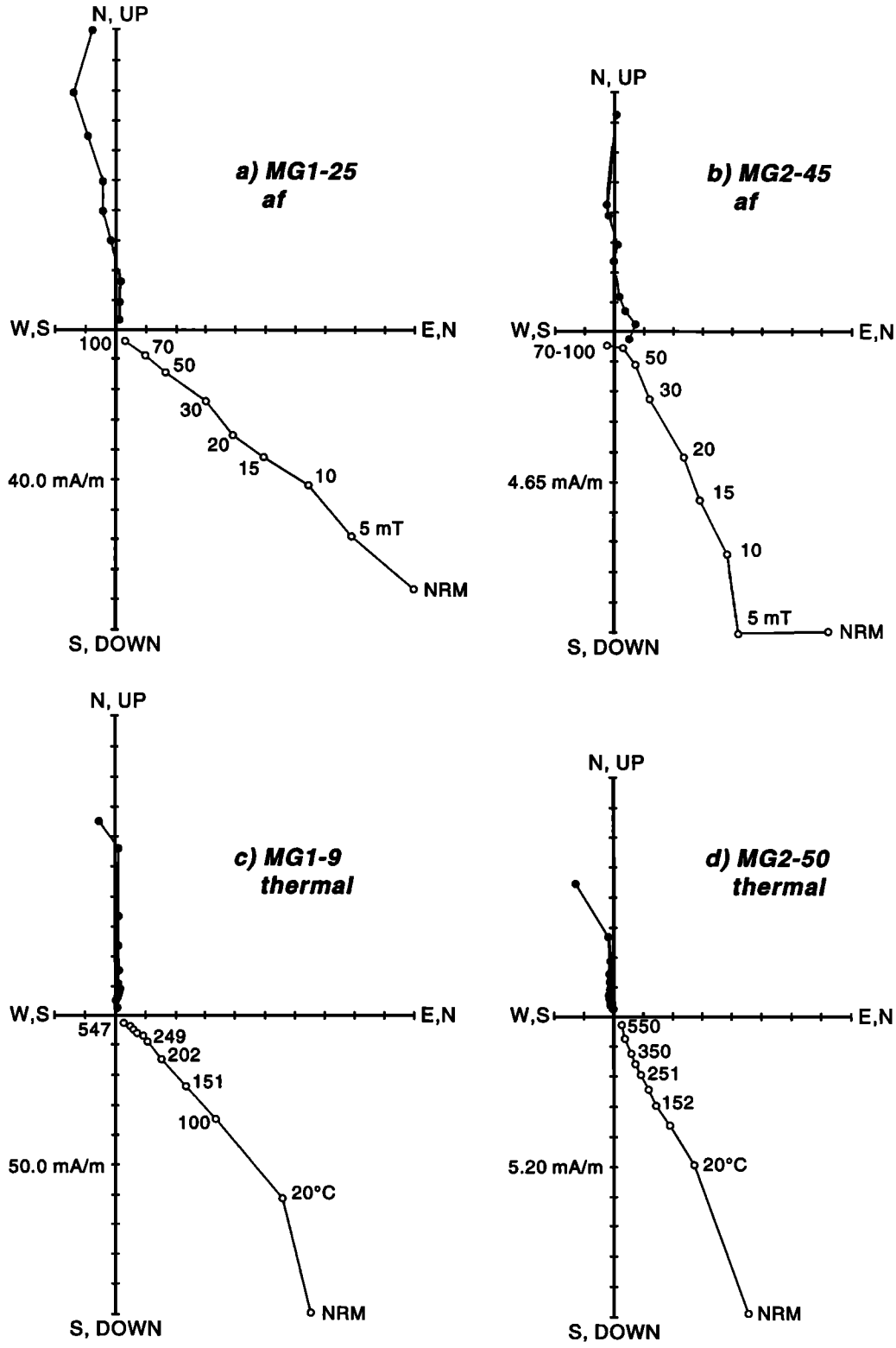


Figure 2. Orthogonal projections of alternating field (af) demagnetization vector endpoints (uncorrected for tilt) for samples (a) MG1-25 and (b) MG2-45, and thermal demagnetization data for samples (c) MG1-9 and (d) MG2-50, showing primarily single-component magnetizations. Higher demagnetization steps (50-100 mT) for MG2-45 might indicate an earlier magnetization now almost completely overprinted (see text). Solid circles indicate projections onto the horizontal plane, and open circles indicate projections onto the vertical plane.

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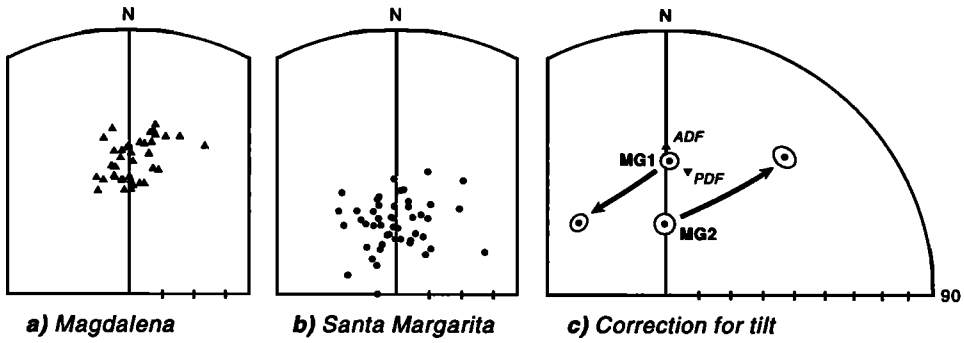


Figure 3. Equal area plots of in situ sample directions for (a) site MG1 on Magdalena Island and (b) site MG2 on Santa Margarita Island. Site-mean directions with 95% confidence limits (Table 1) are plotted both before and after structural correction in Figure 3c. Also included in Figure 3c are the axial dipole field direction (ADF) and the present-day field (PDF) direction (1985.0 IGRF, *Peddie* [1987]) for the Bahia Magdalena area. The site-mean directions fail a fold test at the 99% confidence level, indicating that the remanent magnetization postdates regional folding (see text). Solid symbols indicate lower hemisphere points.

4. Discussion

4.1. Magdalena-Valle Group Comparison

The similarity of results for the Magdalena Group in Baja California Sur with those for the Valle Group on the Vizcaino Peninsula suggests that remagnetization in these rocks is more common than previously thought. Although *Hagstrum et al.* [1985] interpreted the characteristic directions for the Valle Group and Upper Cretaceous La Bocana Roja Formation at some localities as primary, two thirds of the data set (12 out of 18 sites) failed a fold test, indicating later remagnetization. Although the remaining six sites passed a fold test, the age of this deformation is unknown and could be relatively recent. Paleomagnetic data reported for the Valle Group by *Patterson* [1984] also show normal-polarity magnetizations similar to the expected late Cenozoic field direction on the Vizcaino Peninsula, and no field stability tests are available for these results.

Several positive stability tests have been reported for the Valle Group on Cedros Island but are questionable. *Smith and Busby* [1993] reported a "slump" test and a fold test for the middle Cretaceous Pinos Formation on Cedros Island. As pointed out by *Butler and Dickinson* [1995], the slump test is inconclusive because there is no statistical improvement (inclination only) after structural correction of the mean directions for all blocks. In response, *Smith and Busby* [1995] claimed a positive conglomerate test because the in situ data for the slump blocks have random directions. This is only half the test, however, and unslumped Pinos Formation must also be shown to carry a nonrandom characteristic magnetization. The fold test for the Pinos Formation on Cedros Island includes data from the Valle Formation on the Vizcaino Peninsula [*Patterson*, 1984], but the age of deformation and, thus, of magnetization acquisition in the two areas is uncertain.

4.2. Widespread Remagnetization of Cretaceous Strata

Many paleomagnetic studies of other Cretaceous forearc strata in southern California and Baja California have reported magnetization directions that are similar to the expected late Cenozoic field direction for North America, either before or after structural correction (Table 2, Figure 4). Most of these studies found only normal-polarity characteristic magnetizations that may or may not be the result of remagnetization (Table 2). Other studies found normal- and reversed-polarity magnetizations, suggesting that the rocks have not been overprinted by pervasive remagnetizations. Fold or other field stability tests are available for some of these studies but in many cases are inconclusive.

In situ paleomagnetic results for the Campanian Cachuma Formation in southern California (Table 2, site ACC; *Hagstrum et al.* [1996]) and for Cenomanian strata at Figueroa Mountain (Table 2, site FGM; *McWilliams and Howell* [1982]) are of normal polarity and generally similar to the late Cenozoic field direction in southern California. Data for the Upper Cretaceous Jalama Formation on San Miguel Island (Table 2, site JAF; *Champion et al.* [1986]) are similar to the expected late Cenozoic direction after structural correction. Overlying Eocene strata, however, contain normal- and reversed-polarity magnetizations, indicating retention of the primary remanence and that complete remagnetization of the section on San Miguel Island did not occur.

Several Campanian to Maastrichtian sections in southern and Baja California have normal- and reversed-polarity magnetizations (Table 2, entries 9-12), generally pass both reversal and fold tests, and thus probably retain primary magnetizations. Strata from the Santa Ana Mountains [*Fry et al.*, 1985] can be correlated with the marine magnetic anomaly Chron 33r and a reversed magnetozone that has been identified in strata of the Great Valley Group in northern California [*Ward et al.*, 1983]. Sections of Upper Jurassic through mid-Cretaceous Great Valley Group along the western margin of the Sacramento Valley, on the other hand, have been remagnetized in a normal-polarity field with steep inclinations, presumably during the Late Cretaceous [*Mankinen*, 1978]. Strata from the Point Loma Formation [*Bannon et al.*, 1989] can be correlated with Chron 32r, and those from the El Rosario Formation can be correlated with Chron 33n [*Filmer and Kirshvink*, 1989]. *Filmer and Kirschvink* [1989] also noted biochemical observations of ammonite shell matrix that indicate virtually no chemical or thermal alteration of the El Rosario Formation.

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Table 2. Paleomagnetic Data for Mesozoic Rocks Along the California Margin

Unit/Locality	λ_S	ϕ_S	I_t	D_t	I_c	D_c	N	R	k	α_{95}	λ_P	ϕ_P	Ref.
Cretaceous Forearc Strata													
Late Cenozoic remagnetization (?)													
Magdalena Group (MG1)*	24.6	247.9	47.7	1.5	54.1	311.6	39	38.6130	98	2.3	86.3	268.8	1
Magdalena Group (MG2)*	24.5	248.1	67.3	0.0	31.3	40.1	48	47.0749	51	2.9	64.4	248.1	1
Valle Group (Arroyo La Pitahaya)	26.9	246.3			36.5	355.6	60	58.2571	35	3.2	82.2	98.4	2
Valle Group (Arroyo San Lorenzo)	27.0	246.2			43.5	352.3	97	93.3704	27	2.8	82.9	145.0	2
Valle Group (Cañon Malarrimo)	27.7	245.6			39.7	11.8	(3)	2.9927	275	7.5	78.3	356.6	2
Valle Group (Upper Gran Cañon)	28.2	244.8			39.1	347.5	16	15.4333	30	7.3	77.2	129.3	2
Valle Group (San Pablo)*	27.3	245.6	41.1	10.2			(6)	5.9632	136	5.8	80.1	355.7	3
Valle Group (Puerto Escondido)*	27.5	245.3	45.0	53.4	29.1	18.0	5	4.9372	64	9.7	69.4	7.3	3
Valle Group (Puerto Nuevo)*	27.5	245.4	39.2	13.2			(6)	5.8869	44	10.2	76.9	356.3	3
Valle Group (Campito)	27.8	245.2	65.6	334.0	46.8	5.1	12	11.9537	237	2.8	85.5	331.5	3
La Bocana Roja Formation	30.1	244.3	58.2	358.7	52.1	12.0	(3)	2.9967	598	5.1	79.5	316.6	3
Valle Group (VG1)	28.2	244.8	-20	359	37	357	(6)	5.8871	62	10	82	85	4
Valle Group (VG2)	28.2	244.8	-20	15	38	10	(4)	3.9606	127	11	79	10	4
Pinos Formation	28.2	244.8	52	3	39	348	8	7.7429	35	11	78	128	4
Jalama Formation	34.0	239.6			43.9	24.0	94	91.0617	32	2.6	67.7	345.2	5
Cachuma Formation	34.5	240.4	58.6	13.9			21	20.8564	139	2.7	77.9	303.2	6
Figueroa Mountain	34.7	240.0	74	2			8	7.7787	32	10	65	242	7
Jacoby Butte*	42.3	235.8	67.0	5.3	66.6	327.0	(6)	5.9074	54	9.2	81.8	260.5	8
Wilber Springs sandstone*	39.0	237.5	66.9	19.1	61.0	68.0	(4)	3.9319	44	14.0	72.8	283.5	8
Wilber Springs calcareous concretions 1*	39.0	237.5	63.3	8.2	36.3	48.4	(6)	5.9297	75	8	81.6	281.1	8
Wilber Springs calcareous concretions 2*	39.0	237.5	42.7	310.6	55.7	359.0	(6)	5.9113	61	9	87.1	73.8	8
Cretaceous, Jurassic, and Paleocene Forearc Strata													
Primary magnetization (?)													
Punta Baja/Rosario Formations (K)	30.0	244.3	46.2	354.8			(17)	14.9887	8	13.5	84.8	127.2	9
Bedford Canyon Formation (Jr)	33.7	242.3	28.1	16.6			(4)	3.8575	21	20.5	66.0	19.6	9
Silverado Formation (Paleocene)	33.7	242.3	45.2	23.4			(4)	3.9689	97	9.4	68.7	345.1	9
Tuna Canyon Formation (K)	34.1	241.4	37.2	26.4			(6)	5.8931	47	9.9	61.8	144.7	9
Point Loma Formation (K)	32.7	242.8			38.5	30.4	26	24.8520	22	6.2	60.9	347.5	10
Ladd and Williams Formations (K)	33.7	242.5	44.6	5.7			(20)	18.5991	14	9.2	81.1	27.5	11
El Rosario Formation (Punta Baja)	30.0	244.2			-47.7	182.6	32	23.7500	4	15.0	87.5	0.4	12
El Rosario Formation (Punta San José)	31.4	243.4			43.3	3.7	99	95.1538	26	3.1	83.0	34.8	12
Coast Range Ophiolite (CRO) and Franciscan Complex (FC) Blocks													
Late Cenozoic remagnetization (?)													
Stanley Mountain chert (CRO)*	35.1	239.7	56.4	8.2			12	11.9139	128	3.9	83.1	311.8	13
Mount Diablo diabase (CRO)*	37.9	238.1	63.4	358.4			47	46.0803	50	3.0	82.9	228.7	14
Del Puerto chert (CRO)*	37.4	238.7	62.0	9.1	52.0	181.1	9	8.8962	78	5.9	80.9	285.7	8
Redrock Canyon chert (FC)*	34.6	240.3	62.5	359.9			6	5.8360	30	12.3	80.8	239.7	6
Goat Rock chert (FC)*	34.7	240.1	65.6	7.0			12	11.8349	67	5.4	75.9	259.8	6
Mount Umunhum chert (FC)*	37.2	238.1	45.5	1.6			11	10.8282	58	6.0	79.7	49.9	15
Marin Headlands chert (FC)*	37.8	237.5	52.8	9.8			26	24.6926	19	6.7	80.9	353.6	15
The Geysers chert (FC)*	38.8	237.2	59.5	352.8			12	11.8255	63	5.5	84.3	165.0	15
The Geysers basalt (FC)*	38.8	237.2	63.1	3.5			9	8.8574	56	6.9	83.6	259.9	15
Laytonville Limestone (FC)*	42.5	236.6	69.5	354.2			(9)	8.9246	132	4.5	78.6	218.8	16
Cretaceous Forearc Strata and Coast Range Ophiolite													
Late Cretaceous remagnetization (?)													
Valle Group (El Plallon)	27.7	245.1			51.3	319.8	30	28.1765	17	6.8	55.1	172.1	2
Great Valley Group*	40.0	246.6	69	356			(6)	5.9771	218	5	77	236	17, 18
Stonyford volcanics (CRO)	39.3	237.5	80.8	347.2	52.1	42.8	(7)	6.9541	133	5.3	56.6	230.4	8
Stonyford sediments (CRO)	39.3	237.5	67.0	323.3	45.7	49.7	(14)	13.2537	17	9.8	62.2	181.5	8
Paskenta (CRO)	39.9	237.3	58.3	327.5	30.2	55.6	11	10.4876	19	10.6	65.0	155.8	8
Eugenia Formation basalt (CRO)*	27.7	245.2	58.0	341.9			(8)	7.6562	20	12.6	71.3	195.8	19
Cretaceous Plutonic Rocks													
Southern California batholith	33.5	243.0					(32)	31.4402	55	3.4	84.2	1.9	3, 20
Peninsular Ranges batholith	27.0	247.5					(7)	6.8875	53	8.3	81.7	285.5	3
San Telmo pluton	31.1	243.9					(15)	14.8730	110	3.7	75.2	170.9	21

Here, λ_S and ϕ_S are north latitude and east longitude of sampling locality; I_t and D_t are in situ inclination and declination of paleomagnetic directions in degrees (I_c, D_c, corrected for structrual tilt; dip is 90° clockwise of strike); N is number of samples (sites); R is vector sum of N unit vectors; k is the concentration parameter [*Fisher*, 1953]; α_{95} is the radius of 95% confidence in degrees; λ_P and ϕ_P are north latitude and east longitude of virtual geomagnetic pole in degrees. Ref., references: 1, this study; 2, *Patterson* [1984]; 3, *Hagstrum et al.* [1985]; 4, *Smith and Busby* [1993]; 5, *Champion et al.* [1986]; 6, *Hagstrum et al.* [1996]; 7, *McWilliams and Howell* [1982]; 8, *Frei and Blake* [1987]; 9, *Morris et al.* [1986]; 10, *Bannon et al.* [1989]; 11, *Fry et al.* [1985]; 12, *Filmer and Kirschvink* [1989]; 13, *Hagstrum and Murchey* [1996]; 14, *Hagstrum and Jones* [1998]; 15, *Hagstrum and Murchey* [1993]; 16, *Tarduno et al.* [1990]; 17, *Mankinen* [1978]; 18, *Mankinen and Irwin* [1982]; 19, *Hagstrum et al.* [1993]; 20, *Teissere and Beck* [1973]; 21, H. Böhnel and L.A. Delgado-Argote (Paleomagnetic data from northern Baja California (Mexico): Results from the Cretaceous San Telmo batholith, unpublished manuscript, 1998).

*Sections with magnetizations that fail fold tests or are clearly secondary.

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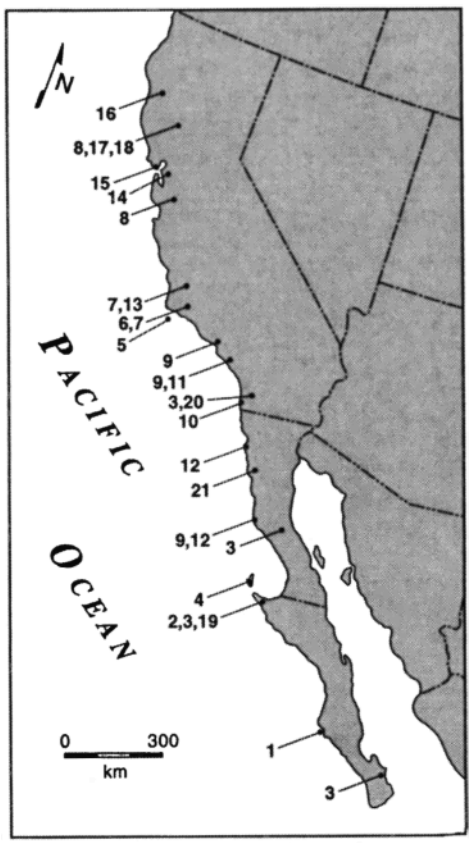


Figure 4. Map of western North America showing localities of the paleomagnetic studies listed in Table 2.

The primary magnetizations in these Upper Cretaceous sections, however, may be of limited utility in tectonic applications because of compaction shallowing of the recorded inclination. *Butler et al.* [1991] argued that the expected offset from compaction shallowing, on the basis of marine sediment core and experimental observations, would be 8°-10°, or most of the 10°-15° inclination anomaly observed in the paleomagnetic data from the sedimentary rocks of southern California and Baja California. *Champion et al.* [1984] sampled the Campanian to Maastrichtian Pigeon Point Formation near Pigeon Point and Paleocene sedimentary rocks at Point San Pedro in northern California and measured paleomagnetic inclinations indicating ~25° of northward transport. More recently, *Kodama and Davi* [1995] disaggregated samples from the Pigeon Point Formation and compacted them under laboratory-simulated field conditions. Their results suggest a ~10° compaction inclination shallowing for the Pigeon Point turbidites and that they instead underwent approximately 13°-16° of northward displacement.

Tan and Kodama [1998] have also measured the magnetic properties of artificially compacted samples of the Ladd Formation and Point Loma Formation in southern California and inferred that these rocks have experienced ~12° and ~17° of inclination shallowing due to burial compaction, respectively. No control experiment using these techniques, however, has been conducted on similar undisplaced rocks to demonstrate that these procedures are not manufacturing the compaction shallowing problem. Furthermore, *Ward et al.* [1997] compared paleomagnetic directions from early cemented carbonate concretions (having undeformed fossils at their center) with those from the siltstones and claystones of the Upper Cretaceous Nanaimo Group on Vancouver Island and found no difference. Similarly, roughly half of the samples collected by *Filmer and Kirschvink* [1989] in the El Rosario Formation of Baja California were from early diagenetic concretions, making inclination shallowing unlikely in these strata as well. In addition, compaction shallowing does not explain the declination anomaly (usually 20° or more of clockwise rotation) determined in most studies from peninsular California. Because several newer data points are inconsistent with the mean rotation, *Butler et al.* [1991] suggest that the data set of primary magnetizations from Upper Cretaceous sections has "deteriorated significantly." Such variability, however, could also be explained by local vertical-axis rotations during subsequent transform-margin tectonism.

4.3. Remagnetization of the CRO, GVG, and Franciscan Complex

Remagnetization directions similar to the expected late Cenozoic field direction have also been reported by several paleomagnetic studies of the Coast Range ophiolite (CRO), overlying Great Valley Group (GVG), and Franciscan Complex of California (Table 2, Figure 5). *Frei and Blake* [1987] sampled Coast Range ophiolite and overlying Great Valley Group rocks or their equivalents at five localities in northern California and Oregon. At Del Puerto (CRO), Wilber Springs (GVG), and Jacoby Butte (GVG), magnetizations with late Tertiary directions postdate the local deformation. At Stonyford and Paskenta (both CRO), however, magnetizations with directions similar to the expected Late Cretaceous field direction predate some of the deformation in these rocks. *Frei and Blake* [1987] attributed the remagnetization events to burial of these rocks to about 7 km during the Cretaceous and tectonic uplift in either the Late Cretaceous or late Tertiary. A secondary magnetization within the Coast Range ophiolite at Mount Diablo in northern California also has been attributed to Pliocene uplift of the Coast Ranges [*Mankinen et al.*, 1991; *Hagstrum and Jones*, 1998].

Blocks of red radiolarian chert within the Franciscan Complex carry a component of magnetization that is similar to the expected late Cenozoic field direction as well [*Hagstrum and Murchey*, 1993; *Hagstrum et al.*, 1996]; these magnetizations are usually removed during thermal demagnetization by ~300°C. This magnetization is perhaps the result of viscous alignment during exposure to the recent geomagnetic field, although it also could be the result of a thermal event associated with the recent uplift of the California Coast Ranges. A similar component of magnetization we interpret as late Cenozoic (Table 2, Figure 5c) was found in the Upper Cretaceous Laytonville Limestone of northern California by *Tarduno et al.* [1990], which they interpreted as a Late Cretaceous-early Tertiary overprint related to accretion within the subduction zone.

4.4 Regional Implications

Paleomagnetic studies indicate that many rocks of the Great Valley Group, the Coast Range ophiolite, and the Franciscan Complex from the Coast Ranges of California and from the

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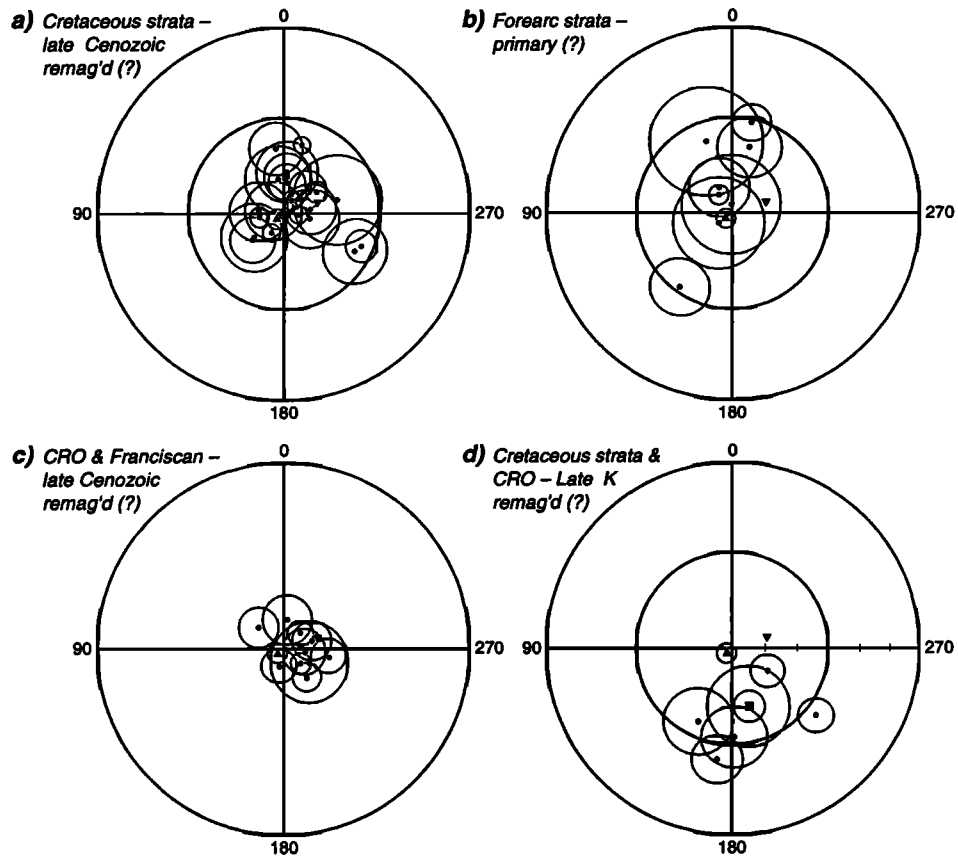


Figure 5. Lambert equal-area projections (30°-90°N latitude) of paleomagnetic pole positions (with 95% confidence limits; averaged *dp* plus *dm* values) for uplifted rocks along the California margin as grouped in Table 2. Triangle (point up) indicates an average Miocene pole position for stable North America [after *Hagstrum et al.*, 1987], the inverted triangle (point down) indicates the pole position of the present-day axial dipole field based on epoch 1985.0 IGRF spherical harmonic coefficients [*Peddie*, 1987], and the square in Figure 5d indicates the Cretaceous pole for North America [*Globerman and Irving*, 1988].

Baja California Peninsula have been magnetically overprinted during the late Mesozoic and Cenozoic. The Magdalena Group in Baja California Sur has clearly been remagnetized after being structurally deformed during the late Tertiary, as have similar forearc strata of the Valle Group on the Vizcaino Peninsula and in southern California. Some of these rocks also pass fold tests, but the age of deformation is generally unknown and magnetization acquisition therefore could have occurred at almost any time. Other Campanian and Maastrichtian forearc strata in southern and Baja California have similar in situ directions but pass fold and reversal tests. Interpretation of the characteristic direction of magnetization within these sedimentary rocks therefore is complicated because it can be attributed to a recent remagnetization, or it may be a primary magnetization that has been affected by either tectonic transport from the south or compaction shallowing of the inclination.

Within the overprinted rocks of the Cretaceous forearc strata, there appear to be two episodes of remagnetization: one in the Late Cretaceous and a second in the late Tertiary. The first group includes localities in the Great Valley Sequence along the western margin of the Sacramento Valley [*Mankinen*, 1978] and the Coast Range ophiolite at Stonyford and Paskenta [*Frei and Blake*, 1987] in northern California (Figure 4), and Jurassic pillow basalts of the Eugenia Formation [*Hagstrum et al.*, 1993] and a site in the Valle Group [*Patterson*, 1984] on the Vizcaino Peninsula. The second group (Table 2, Figure 5) includes the Magdalena Group, the Valle Group, and other forearc strata in southern California, and the Coast Range ophiolite and Franciscan Complex in northern and southern California. Remagnetization at each locality may have been caused by events of local areal extent, including Cenozoic volcanism, chemical alteration, and viscous overprinting. However, the widespread distribution of units with similar paleomagnetic results may indicate that remagnetization was caused by events of regional scope.

Regionally extensive remagnetization may have been a result of widespread exhumation, surface uplift, or both [*Pullaiah et al.*, 1975]. Western California and Baja California underwent extensive Late Cretaceous to early Paleogene tectonism related to the exhumation of blueschists [*Blake et al.*, 1988; *Sedlock*, 1996, 1998], and surface uplift is implied by the presence of Franciscan detritus in early Tertiary conglomerate near Stonyford [*Dickinson et al.*, 1979]. Exhumation has been attributed to underplating and syn-subduction extension [*Sedlock*, 1996, 1998], and surface uplift has been related to

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Table 3. Reference Poles and Displacement Parameters for Southern and Baja California

Unit	λ_R	ϕ_R	A_{95}	d	r	Ref.
Forearc Strata						
Silverado Formation (Paleocene)	81.5	192.6	3.2	12±6	32±11	9
Punta Baja/Rosario Formations (K)	71.1	195.7	4.9	14±10	14±16	9
El Rosario Formation (K)	71.1	195.7	4.9	16±4	24±6	12
Point Loma Formation (K)*	71.1	195.7	4.9	23±5	50±8	10
Ladd and Williams Formations (K)*	71.1	195.7	4.9	19±7	25±12	11
Tuna Canyon Formation (K)	71.1	195.7	4.9	19±7	-12±12†	9
Bedford Canyon Formation (Jurassic)	70.2	148.8	6.7	15±11	40±19	9
Cretaceous Plutonic Rocks						
Southern California batholith	71.1	195.7	4.9	14±5	26±6	3, 20
Peninsular Ranges batholith	71.1	195.7	4.9	4±8	25±9	3
San Telmo pluton	71.1	195.7	4.9	8±5	2±6	21

Here, λ_R and ϕ_R are latitude and east longitude of North American reference pole in degrees; A_{95} is radius of 95% confidence of reference pole in degrees; Paleocene pole from *Diehl et al.* [1983]; Cretaceous pole from *Globerman and Irving* [1988]; Upper Jurassic (150 Ma) pole from *Besse and Courtillot* [1991]; d and r are poleward displacement and clockwise rotation of site relative to stable North America in degrees. Ref.: references as in Table 2.

*Samples from these formations subjected to compaction shallowing experiments [*Tan and Kodama*, 1998].

†This study was done in the Western Transverse Ranges Miocene rotated province and cannot be adequately corrected for that rotational effect.

rapid convergence of young Farallon lithosphere [*Molnar and Atwater*, 1978] or to transpressional passage of northward translated terranes [*Maxson and Tikoff*, 1996]. Late Cenozoic surface uplift of the western margin of North America is indicated by Pliocene and younger shortening and uplift in the Coast Ranges [*Page and Engebretson*, 1984], and abundant blueschist detritus in late Cenozoic deposits [*Blake et al.*, 1984; *Kilmer*, 1977]. Shortening and uplift may have occurred in reponse to changes in relative plate motion between the Pacific plate and western North America during 3.4 and 3.9 Ma [*Harbert*, 1991].

If burial and uplift caused remagnetization of forearc strata within the Great Valley Group, then preservation of primary magnetizations in the Campanian and Maastrichtian strata toward the top of the section would be expected. The results deemed primary in this study are listed in Table 3. Paleomagnetic data from plutonic rocks, especially from the large number of sites in southern California, show similar results, but *Butler et al.* [1991] have suggested that these localities are tilted. A more recent paleobarometric analysis of the Peninsular Ranges batholith by *Ague and Brandon* [1992], however, indicates northward transport (700 ± 450 km) but no significant clockwise rotation.

Not all sedimentary rocks show evidence of compaction inclination shallowing [e.g., *Ward et al.*, 1997], and the agreement in amount of northward translation between the forearc strata (average d: 17°±3°) and the southern California batholith (14°±5°) is most simply interpreted as northward translation (14°±3°) and clockwise rotation (29°±8°) of the Peninsular Ranges terrane. Northward transport (up to 4000 km) along the North American margin since the Cenomanian is also indicated by reliable paleomagnetic data (positive fold and reversal tests) from red (hematite-bearing) Franciscan-type chert [*Hagstrum and Sedlock*, 1992; *Hagstrum and Murchey*, 1993; *Hagstrum et al.*, 1996]. Both the primary paleomagnetic data and plate reconstruction models [*Engebretson et al.*, 1985] indicate that these rocks were accreted to the North American margin at paleolatitudes corresponding to what is now southern Mexico. Mesozoic subduction complex rocks are found along the continental margin from Alaska to Baja California Sur and along the Central American margin in Costa Rica and Panama. These rocks are missing from the southern margin of Mexico, consistent with tectonic removal of a portion of this margin by right-lateral transport sometime between Late Cretaceous and late Miocene time [*Karig et al.*, 1978].

5. Summary

1. Cretaceous forearc strata on Magdalena and Santa Margarita Islands, southwestern Baja California, acquired their characteristic magnetization after major tilting of Late Cretaceous to Paleogene age. Minor tilting after remagnetization probably occurred in the late Cenozoic.

2. Other Cretaceous forearc strata throughout southern and Baja California also have characteristic magnetization directions similar to the expected late Cenozoic field direction for North America. These data can be divided into three groups: (1) remagnetized sections failing fold tests, (2) sections with magnetizations of one polarity and no or inconclusive fold tests (possibly remagnetized), and (3) sections with both normal- and reversed-polarity magnetizations (likely primary).

3. The late Cenozoic overprint direction is similar to the primary Cretaceous direction, and these results for southern and Baja California have been interpreted as resulting from either northward tectonic transport and clockwise rotation, or compaction shallowing in sedimentary rocks and tilting in plutonic rocks. The >20° clockwise rotation observed in most studies is not explained by compaction shallowing.

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4. Remagnetization is widespread in the Great Valley Group, Coast Range ophiolite, and Franciscan Complex along the California margin. Apparently, some units were remagnetized in the Late Cretaceous, and others were remagnetized in the late Cenozoic.

5. Regionally extensive remagnetization may have resulted from Late Cretaceous to Paleogene exhumation and to late Cenozoic shortening and surface uplift. Local causes, however, could include reheating due to volcanism, chemical alteration, or viscous overprinting.

6. Remagnetization must be regarded as a major problem in rocks that have been subjected to accretion, burial, and uplift along the California and other convergent margins.

7. We prefer to interpret the coincidence between the results for the southern California batholith and forearc strata of southern and Baja California as indicating 14°±3° of northward transport and 29°±8° of clockwise rotation. Presently, too few data are available from the Peninsular Ranges batholith in Mexico, and more are needed to test the hypotheses of northward transport and tectonic tilt.

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